

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-2/T-2 CSE 219: Signals and Linear Systems

Time: 25 minutes

Marks: 20

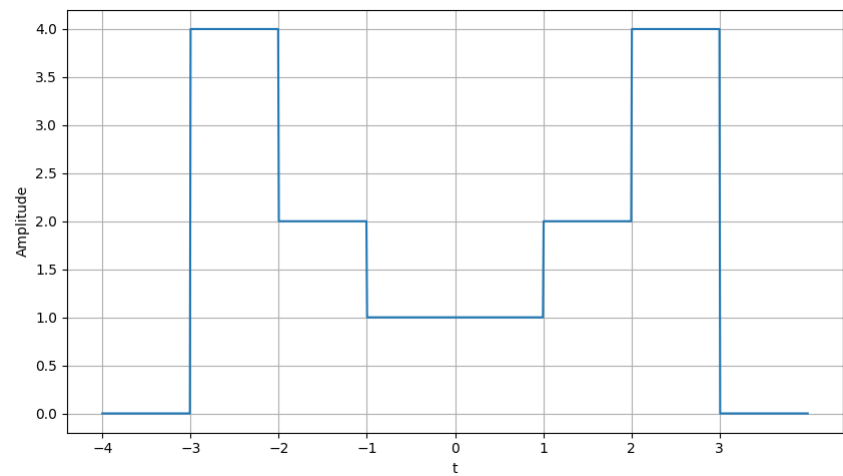
Student Name: _____

Student No: _____

1. Find fourier transform of the signal $x(t) = te^{-2t}u(t - 3)$

(8 Marks)

2. Use linearity property to find fourier transform of the signal given in the plot below. You are not allowed to use the fourier transform formula directly. (7 Marks)



3. Given a signal $x(t)$, outline the steps for computing its second derivative $x''(t)$ using Fourier Transform and Inverse Fourier Transform. (5 Marks)